

LECTURE NOTES FOR SURV 616/686

Statistical Methods II

Categorical Data Analysis

CLASS #09

OVERVIEW: In this class we will introduce the basic sampling distributions, and their properties, for discrete categorical data. We will also begin looking at goodness of fit statistics, and maximum likelihood methods for the basic sampling distributions.

References

Agresti, Alan (1990). Categorical Data Analysis. John Wiley and Sons.

Bishop, Fienberg, and Holland (1975). Discrete Multivariate Analysis. MIT Press.

Fienberg, Stephen E. (1980). The Analysis of Cross-Classified Categorical Data. MIT Press.

1. Basic Sampling Distributions for Categorical Data

In this section we will introduce some basic sampling distributions used in categorical data analysis.

The Bernoulli Distribution

Let the random variable X take on two values, 0 and 1, with the following distribution

$$(1.1) \quad P\{X = 1\} = p$$

$$P\{X = 0\} = 1 - p$$

then X is called a *Bernoulli random variable*. Bernoulli random variables are fundamental to the theory of statistics because they form the basic groundwork of the formation of other random variables. We will use the notation $X \sim \text{Ber}(p)$ to denote that the random variable X has a Bernoulli distribution with parameter p . Note that if $X \sim \text{Ber}(p)$ then

$$(1.2) \quad E\{X\} = p$$

$$V\{X\} = p(1-p)$$

The Binomial Distribution

Let $X_1, \dots, X_n$ be a sample of independent $\text{Ber}(p)$ random variables, and define

$$(1.3) \quad Y = \sum_{i=1}^n X_i$$

then we can write *probability mass function* for Y as

$$(1.4) \quad P\{Y = y\} = \frac{n!}{y!(n-y)!} p^y (1-p)^{n-y} \quad \text{for } y = 0, \dots, n$$

The random variable Y is said to have a *Binomial distribution* with parameters n and p which we write $Y \sim \text{Bin}(n, p)$. The key assumptions for a Binomial random variable are

Assumptions for Binomial Distribution

1. n Independent trials
2. Only two possible outcomes on each trial
3. The probability of “success” p is the same for each trial

These assumptions correspond to simple random sampling with *replacement*. However, it is a good approximation to sampling *without replacement* from large populations with

small sampling fractions. The mean and variance of a random variable with a binomial distribution is given by

$$(1.5) \quad E\{Y\} = np$$

$$V\{Y\} = np(1-p)$$

For a binomial random variable the sample size n is given, and the probability p is generally unknown, and we wish to make inference about the parameter p . We can write the *likelihood function* for a random variable with a binomial distribution as

$$(1.6) \quad L(p; Y, n) = \frac{n!}{Y!(n-Y)!} p^Y (1-p)^{n-Y}$$

and the *log-likelihood function* as

$$(1.7) \quad \ln[L(p; Y, n)] = \ln\left[\frac{n!}{Y!(n-Y)!}\right] + Y \ln(p) + (n-Y) \ln(1-p)$$

The maximum likelihood estimate for p is obtained as the solution to

$$(1.7) \quad \frac{\partial}{\partial p} \ln[L(p; Y, n)] = 0$$

which implies

$$(1.8) \quad \frac{Y}{\hat{p}} - \frac{n-Y}{1-\hat{p}} = 0$$

$$Y(1-\hat{p}) = \hat{p}(n-Y)$$

$$Y = n\hat{p} \Rightarrow \hat{p} = n^{-1}Y$$

It can be shown that the likelihood function in (1.6) is unimodal, which means that the solution obtained in (1.8) is unique. We next give the properties of $\hat{p}$.

Properties of $\hat{p}$ from a Binomial Distribution

1. $\hat{p}$ is a random variable
2. $\hat{p}$ is unbiased, $E\{\hat{p}\} = p$.
3. $V\{\hat{p}\} = n^{-1}p(1-p)$
4. For large n , $\sqrt{n}(\hat{p} - p) \sim N[0, p(1-p)]$ (approximately)
5. The M.L.E. for any one-to-one function of p , say $\theta = g(p)$, is $\hat{\theta} = g(\hat{p})$.

The Multinomial Distribution

Let $\mathbf{X}$ be a $k \times 1$ dimensional random vector which has all its components equal to zero, except for one which is equal to 1. Further we assume that

$$(1.9) \quad \mathbf{X}' = (X_1, \dots, X_k)$$

$$P\{X_j = 1\} = p_j \quad \text{for} \quad j = 1, \dots, k \quad \sum_{i=1}^k p_i = 1$$

Then we can write

$$(1.10) \quad E\{X_j\} = p_j$$

$$V\{X_j\} = p_j(1-p_j)$$

$$C\{X_i, X_j\} = -p_i p_j \quad \text{for} \quad i \neq j$$

In matrix notation we can write

$$(1.11) \quad E\{\mathbf{X}\} = \mathbf{p}$$

$$V\{\mathbf{X}\} = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}'$$

where $\text{diag}(\mathbf{p})$ denotes the diagonal matrix with the vector $\mathbf{p}$ on the diagonal.

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be a sample of interdependent random vectors with the distribution described in (1.9)-(1.11), and define

$$(1.12) \quad \mathbf{Y} = \sum_{i=1}^n \mathbf{X}_i$$

Then $\mathbf{Y}$ is said to have a multinomial distribution with parameters n and $\mathbf{p}$. We can the *probability mass function* of $\mathbf{Y}$ as

$$(1.13) \quad P\{\mathbf{Y}=\mathbf{y}\} = \frac{n!}{\prod_{i=1}^k y_i!} \prod_{i=1}^k p_i^{y_i}$$

The key assumptions for a multinomial random variable are

Assumptions for Multinomial Distribution

1. n Independent trials
2. k possible outcomes on each trial.
3. Identical trials, with

$$p_1 = \text{probability of first outcome}$$

$$\vdots$$

$$p_k = \text{probability of } k\text{-th outcome}$$

These assumptions correspond to simple random sampling *with replacement*. However, it is a good approximation to sampling *without replacement* from large populations with small sampling fractions. The mean and variance of a random vector with a multinomial distribution is given by

$$(1.14) \quad E\{\mathbf{Y}\} = n\mathbf{p}$$

$$V\{\mathbf{Y}\} = n[\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}']$$

For a multinomial random variable the sample size n is given, and the parameter vector $\mathbf{p}$ is generally unknown, and we wish to make inference about $\mathbf{p}$. There are two

methods for deriving the maximum likelihood estimator for $\mathbf{p}$. The methods use different ways of incorporating the restriction that $\sum_{i=1}^k p_i = 1$.

Method 1 (for MLE for the multinomial distribution)

Substitution of $p_k = 1 - \sum_{i=1}^{k-1} p_i$

Write $p_k = 1 - \sum_{i=1}^{k-1} p_i$, and write the likelihood function for the remaining parameters as

$$(1.18) \quad L(\mathbf{p}; \mathbf{Y}, n) = \frac{n!}{\prod_{i=1}^k Y_i!} \left(1 - \sum_{i=1}^{k-1} p_i \right)^{Y_k} \prod_{i=1}^{k-1} p_i^{Y_i}$$

and the *log-likelihood function* as

$$(1.19) \quad \ln [L(\mathbf{p}; \mathbf{Y}, n)] = \ln \left(\frac{n!}{\prod_{i=1}^k Y_i!} \right) + Y_k \ln \left(1 - \sum_{i=1}^{k-1} p_i \right) + \sum_{i=1}^{k-1} Y_i \ln (p_i)$$

The maximum likelihood estimator is obtained as the solution to

$$(1.20) \quad \frac{\partial}{\partial p_i} \ln [L(\mathbf{p}; \mathbf{Y}, n)] = 0 \quad \text{for} \quad i = 1, \dots, k-1$$

which implies

$$(1.21) \quad \frac{Y_i}{\hat{p}_i} - \frac{Y_k}{1 - \sum_{j=1}^{k-1} \hat{p}_j} = 0$$

which yields the solution

$$(1.22) \quad \hat{p}_i = n^{-1} Y_i \quad \text{for} \quad i = 1, \dots, k$$

$$\hat{\mathbf{p}} = n^{-1} \mathbf{Y}$$

Method 2 (for MLE for the multinomial distribution)
Lagrange Multiplier Method

Maximize the objective function

$$(1.23) \quad G(\mathbf{p}, \lambda) = \ln [L(\mathbf{p}; \mathbf{Y}, n)] - \lambda \left(\sum_{i=1}^k p_i - 1 \right)$$

Then we get

$$(1.24) \quad \frac{\partial}{\partial p_i} G(\mathbf{p}, \lambda) = \frac{Y_i}{p_i} - \lambda \quad \text{for } i = 1, \dots, k$$

$$\frac{\partial}{\partial \lambda} G(\mathbf{p}, \lambda) = - \left(\sum_{i=1}^k p_i - 1 \right)$$

and by setting the partial derivatives in (1.24) equal to zero we get

$$(1.25) \quad \hat{\mathbf{p}} = n^{-1} \mathbf{Y}$$

$$\hat{\lambda} = n$$

We next give the properties of the MLE $\hat{\mathbf{p}}$.

Properties of $\hat{\mathbf{p}}$ from a Multinomial Distribution

1. $\hat{\mathbf{p}}$ is a random vector
2. $\hat{\mathbf{p}}$ is unbiased, $E\{\hat{\mathbf{p}}\} = \mathbf{p}$
3. $V\{\hat{\mathbf{p}}\} = n^{-1} [\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}']$, $V\{\hat{p}_i\} = n^{-1} p_i (1 - p_i)$, $C\{\hat{p}_i, \hat{p}_j\} = -n^{-1} p_i p_j$
4. For large n , $\sqrt{n}(\hat{\mathbf{p}} - \mathbf{p}) \sim N_k(\mathbf{0}, \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}')$ (approximately).
5. The MLE of a function $\theta = g(\mathbf{p})$, is $\hat{\theta} = g(\hat{\mathbf{p}})$.

A particular application of property 5 above is that the MLE of the covariance

matrix of $\hat{\mathbf{p}}$ is given by

$$(1.26) \quad \hat{V}\{\hat{\mathbf{p}}\} = n^{-1} [\text{diag}(\hat{\mathbf{p}}) - \hat{\mathbf{p}}\hat{\mathbf{p}}']$$

and the MLE of the expected frequency $m_i = E\{Y_i\} = np_i$

$$(1.27) \quad \hat{m}_i = n\hat{p}_i.$$

Some care needs to be taken with the limiting Normal distribution resulting in property 4. The multivariate normal distribution

$$(1.28) \quad \sqrt{n}(\hat{\mathbf{p}} - \mathbf{p}) \sim N_k(\mathbf{0}, \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}')$$

is a special type multivariate normal distribution known as a k dimensional *singular multivariate normal distribution*. The singularity refers to the fact that

$$(1.29) \quad \sum_{i=1}^k \hat{p}_i = 1$$

which says that any one of the elements of $\hat{\mathbf{p}}$, say $\hat{p}_i$, can be written in terms of the remaining elements. Therefore, given the remaining elements, $\hat{p}_i$ is known with certainty and thus has zero variance. This implies that the covariance matrix is *singular* since

$$(1.30) \quad [\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}'] \mathbf{1}_k = \mathbf{p} - \mathbf{p} = \mathbf{0}$$

We have dealt up to this point in SURV 616 with covariance matrices which are nonsingular. The properties of the singular normal distribution are the same as those of the regular multivariate normal distribution, except when the inversion of the covariance matrix is concerned. For example, we can not form that statistic

$$(1.31) \quad (\hat{\mathbf{p}} - \mathbf{p}_0)' (V\{\hat{\mathbf{p}}\})^{-1} (\hat{\mathbf{p}} - \mathbf{p}_0) \quad \text{NOT POSSIBLE}$$

to test the hypothesis that

$$(1.32) \quad H_0 : \mathbf{p} = \mathbf{p}_0 \quad \text{versus} \quad H_A : \mathbf{p} \neq \mathbf{p}_0$$

However we can test the hypothesis in (1.32) by removing the last element of $\mathbf{p}$ and reducing the dimensionality of the problem to $k-1$ from k . We will discuss this more later.

Finally, we have been concentrating on the joint distribution of $Y_1, \dots, Y_k$, but the marginal distributions are also of interest. Under the multinomial distribution we have that the marginal distributions are given by

$$(1.33) \quad \begin{aligned} Y_i &\sim \text{Bin}(n, p_i) \\ E\{Y_i\} &= np_i \\ V\{Y_i\} &= np_i(1 - p_i) \end{aligned}$$

The Poisson Distribution (Poisson Sampling)

Let Y_i be a *Poisson random variable* with parameter m_i for $i = 1, \dots, k$. We will write this as $Y_i \sim \text{Poisson}(m_i)$. Also define the vector $\mathbf{Y}$ as

$$(1.34) \quad \mathbf{Y}' = (Y_1, \dots, Y_k)$$

We can write the probability mass function of Y_i as

$$(1.35) \quad \{Y_i = y_i\} = \frac{e^{-m_i} m_i^{y_i}}{y_i!} \quad \text{for} \quad y_i = 0, 1, \dots, \infty$$

For a Poisson random variable we have

$$(1.36) \quad \begin{aligned} E\{Y_i\} &= m_i \\ V\{Y_i\} &= m_i \end{aligned}$$

If we specify that $Y_1, \dots, Y_k$ are *independent*, then we can write the *joint probability mass function* of the vector $\mathbf{Y}$ as

$$(1.37) \quad P\{\mathbf{Y} = \mathbf{y}\} = \prod_{i=1}^k \frac{e^{-m_i} m_i^{y_i}}{y_i!}$$

Unlike the multinomial distribution where the sample size n was fixed, the sample size

$$(1.38) \quad n = \sum_{i=1}^k Y_i$$

is not fixed with the Poisson sampling. The assumptions are summarized below

Assumptions with Poisson Sampling

1. k Independent categories
2. Expected count for category i is m_i

We are interested in making inference about the parameters $m_1, \dots, m_k$ with Poisson sampling. We can write the log likelihood as

$$(1.39) \quad \ln[L(\mathbf{m}; \mathbf{Y})] = \sum_{i=1}^k Y_i \ln(m_i) - \sum_{i=1}^k m_i - \sum_{i=1}^k \ln(y_i!)$$

and setting

$$(1.40) \quad \frac{\partial}{\partial m_i} \ln[L(\mathbf{m}; \mathbf{Y})] = 0$$

we obtain

$$(1.41) \quad \hat{m}_i = Y_i \quad \text{for} \quad i = 1, \dots, k$$

$$\hat{\mathbf{m}} = \mathbf{Y}$$

We next summarize the properties of $\hat{\mathbf{m}}$ from Poisson sampling.

Properties of $\hat{\mathbf{m}}$ from Poisson Sampling

1. $\hat{\mathbf{m}}$ is a random vector
2. $\hat{\mathbf{m}}$ is unbiased, $E\{\hat{\mathbf{m}}\} = \mathbf{m}$
3. $V\{\hat{\mathbf{m}}\} = \text{diag}(\mathbf{m})$,
4. The MLE of a function $\theta = g(\mathbf{m})$, is $\hat{\theta} = g(\hat{\mathbf{m}})$.

Notice that we did not give a large sample normal distribution result for Poisson sampling as we did for the multinomial distribution. This has to do with the fact that with multinomial sampling for the sample size n is fixed, and we develop a large sample distribution as n gets large. In the Poisson sampling case the sample size is a random variable so it is slightly more difficult to speak of a large sample result in this case. We will return to this subject later.

While we have repeatedly stressed that the sample size n with Poisson sampling is a random variable, it is interesting to consider the sampling distribution of $\mathbf{Y}$ when we condition on the observed sample size n . Remember that the sample size can be written

$$(1.42) \quad n = \sum_{i=1}^k Y_i$$

so from mathematical statistics we know that

$$(1.43) \quad n \sim \text{Poisson}\left(\sum_{i=1}^k m_i\right)$$

where we used the fact that $Y_1, \dots, Y_k$ are independent. Therefore we can write

$$(1.44) \quad P\left\{\mathbf{Y} = \mathbf{y} \mid \sum_{i=1}^k Y_i = n\right\} = \frac{P\left\{\mathbf{Y} = \mathbf{y}, \sum_{i=1}^k Y_i = n\right\}}{P\left\{\sum_{i=1}^k Y_i = n\right\}}$$

$$\begin{aligned}
&= \frac{\prod_{i=1}^k \frac{e^{-m_i} m_i^{y_i}}{y_i!}}{(n!)^{-1} \left(\sum_{i=1}^k m_i \right)^n \exp \left(-\sum_{i=1}^k m_i \right)} = \frac{n!}{\prod_{i=1}^k y_i!} \frac{\prod_{i=1}^k m_i^{y_i}}{\left(\sum_{i=1}^k m_i \right)^n} \\
&= \frac{n!}{\prod_{i=1}^k y_i!} \frac{\prod_{i=1}^k m_i^{y_i}}{\prod_{i=1}^k \left(\sum_{i=1}^k m_i \right)^{y_i}} = \frac{n!}{\prod_{i=1}^k y_i!} \prod_{i=1}^k \left(\frac{m_i}{\sum_{i=1}^k m_i} \right)^{y_i} = \frac{n!}{\prod_{i=1}^k y_i!} \prod_{i=1}^k p_i^{y_i}
\end{aligned}$$

where

$$(1.45) \quad p_i = \frac{m_i}{\sum_{i=1}^k m_i} \quad \text{for} \quad i = 1, \dots, k$$

Therefore Poisson sampling conditional upon the observed total sample size has the same distribution as multinomial sampling with $\mathbf{p}$ defined by (1.45).

2. Notation for Categorical Analysis and Further Comments on Sampling

Up to now we have talked abstractly about some sampling distributions for discrete data. We now want to be more concrete and introduce some general notation. We begin by considering *two-way tables* which will have a particular characteristic associated with the rows of the table and another characteristic associated with the columns of the tables. In general the rows of the table will be indexed by i where $i = 1, \dots, I$ and the columns will be indexed by j where $j = 1, \dots, J$. For the 4×5 case for example we write

(2.1)

Rows	Columns					
	1	2	3	4	5	
1	n_{11}	n_{12}	n_{13}	n_{14}	n_{15}	n_{1+}
2	n_{21}	n_{22}	n_{23}	n_{24}	n_{25}	n_{2+}
3	n_{31}	n_{32}	n_{33}	n_{34}	n_{35}	n_{3+}
4	n_{41}	n_{42}	n_{43}	n_{44}	n_{45}	n_{4+}
	n_{+1}	n_{+2}	n_{+3}	n_{+4}	n_{+5}	n

where n_{ij} is the sample cell count and

$$(2.2) \quad n_{i+} = \sum_{j=1}^J n_{ij}$$

$$n_{+j} = \sum_{i=1}^I n_{ij}$$

$$n = \sum_{i=1}^I n_{i+} = \sum_{j=1}^J n_{+j} = \sum_{i=1}^I \sum_{j=1}^J n_{ij}$$

Tables of counts can be deceptively simple in appearance because they do not reveal the way in which the data was collected, and in particular the sampling distribution we should use to describe the cell counts. We will generally consider the following three scenarios.

Multinomial Sampling

In this case we consider n repeated independent trials with $I \times J$ categories. For example we select a random sample of individuals and then classify them by their row and column attributes and place them in the appropriate one of the $I \times J$ categories. Tables derived from this form of sampling are sometimes called *cross-sectional* studies.

Product (or Independent) Multinomial Sampling

In this case we take J independent samples of individuals. In the first sample, of size n_{+1} , we select a random sample of individuals who are known to have the attribute of Column 1, and then classify them into the appropriate row. Likewise in the j -th

sample, of size n_{+j} , we select a random sample of individuals who are known to have the attribute of column j , and then classify them into the appropriate row. In this way we have a table composed of J independent multinomial samples each with I categories but with (generally) different probability vectors $\mathbf{p}_j$ where $\mathbf{p}_j$ is $I \times 1$.

We have given the example above in terms of sampling within columns, but the sampling can be thought of as being done within rows instead. In some cases we can think of the columns as corresponding to a response variable, and the rows as corresponding to an independent variable (an illustration of this is given in the next example). Then if the independent sampling is done within each of the categories of the independent variable then this is called a *prospective* study since we are controlling the totals for each of the categories of the independent variable. Conversely, if the independent sampling is done within each of the categories of the response variable then this is called a *retrospective* study since we are controlling the totals for each of the categories of the response variable.

Poisson Sampling

In this case we treat the (i, j) cell in the table as an independent Poisson random variable with parameter m_{ij}

We next present a hypothetical example.

Example 1. This example is taken from Categorical Data Analysis by Alan Agresti. Suppose that researchers in the Massachusetts Department of Transportation plan to study the relationship between seat-belt use (yes, no) and outcome of crash (fatality, non-fatality), for drivers involved in automobile accidents on the Massachusetts Turnpike. They will summarize results in the table below.

<i>Seat-Belt Use and Results of Automobile Crashes</i>		
<i>Seat-Belt use</i>	<i>Result of Crash</i>	
	<i>Fatality</i>	<i>Nonfatality</i>
<i>Yes</i>	n_{11}	n_{12}
<i>No</i>	n_{21}	n_{22}

Suppose they catalog all accidents on the Turnpike for the next year, classifying each according to these variables. Then the total sample size is a random variable, and they might treat the numbers of observations at the four combinations of seat-belt use and outcome of crash as independent Poisson random variables with unknown means $m_{11}, m_{12}, m_{21}, m_{22}$.

Suppose, instead, the researchers randomly sample 100 police records of crashes on the Turnpike in the last year, and classify each according to seat-belt use and outcome of crash. For this cross-sectional study, the total sample size n is fixed, and they might treat the four cell counts as a multinomial random variable with $n = 100$ trials and unknown probabilities $\pi_{11}, \pi_{12}, \pi_{21}, \pi_{22}$.

Suppose police records for accidents involving fatalities were filed separately from the others. The researchers might instead randomly sample 50 records of accidents with a fatality and 50 records of accidents with no fatality. For this retrospective approach, the column totals and the table are fixed. They might then regard each column as an independent binomial (multinomial with two levels) sample.

Note that the last situation of the example was called *retrospective* since the Result of Crash is being considered the response variable, and Seat-Belt Use is considered an independent variable. If the police classified all accidents by seat-belt use, then we could take sample of 50 accidents with a seat-belt, and 50 accidents without a seat-belt. This would then be a prospective approach where the row totals are fixed. Then each row would be considered an independent binomial sample.

3. Testing Goodness of Fit with Specified Multinomial Probabilities

In this section we consider the following important case. Let $(n_1, \dots, n_k)$ denote the observed responses on a multinomial random vector with parameters n and $\mathbf{p}' = (p_1, \dots, p_k)$, where $\sum_{i=1}^k n_i = n$. Imagine that we want to test the hypothesis that

$$(3.1) \quad H_0 : \mathbf{p} = \mathbf{p}_0 \quad \text{versus} \quad H_A : \mathbf{p} \neq \mathbf{p}_0$$

where $\mathbf{p}'_0 = (p_{01}, \dots, p_{0k})$. We present two ways of testing this hypothesis.

The Pearson Chi-Square Statistic X^2

We know that

$$(3.2) \quad \hat{\mathbf{p}}' = (\hat{p}_1, \dots, \hat{p}_k) = \left(\frac{n_1}{n}, \dots, \frac{n_k}{n} \right)$$

and under the null hypothesis

$$(3.3) \quad V\{\hat{\mathbf{p}}\} = n^{-1} \left[\text{diag}(\mathbf{p}_0) - \mathbf{p}_0 \mathbf{p}_0' \right]$$

We saw earlier in the discussion about expression (1.31) that we cannot invert the covariance matrix in (3.3) so the natural Wald statistic to test the hypothesis can not be used. Instead, define the $(k-1) \times 1$ vectors

$$(3.4) \quad \begin{aligned} \theta_0' &= (p_{01}, p_{02}, \dots, p_{0,k-1}) \\ \hat{\theta}' &= (\hat{p}_1, \hat{p}_2, \dots, \hat{p}_{k-1}) \end{aligned}$$

and define the $(k-1) \times (k-1)$ matrix under the null hypothesis in (3.1) as

$$(3.5). \quad \Sigma_{\theta} = V\{\hat{\theta}\}$$

where

$$(3.6) \quad \begin{aligned} (\Sigma_{\theta})_{ii} &= n^{-1} p_{0i} (1 - p_{0i}) \\ (\Sigma_{\theta})_{ij} &= -n^{-1} p_{0i} p_{0j} \quad \text{for } i \neq j \end{aligned}$$

It can be shown that the matrix Σ_{θ} is invertible so we can form the Wald statistic

$$(3.7) \quad W = (\hat{\theta} - \theta_0)' \Sigma_{\theta}^{-1} (\hat{\theta} - \theta_0) \sim \chi_{k-1}^2$$

where the distributional result holds for large samples under the null hypothesis in (3.1).

It can be shown that

$$(3.8) \quad \left(\Sigma_{\theta}^{-1}\right)_{ii} = n\left(p_{0i}^{-1} + p_{0k}^{-1}\right)$$

$$\left(\Sigma_{\theta}^{-1}\right)_{ij} = \frac{n}{p_{0k}} \quad \text{for} \quad i \neq j$$

We can further write

$$(3.9) \quad \Sigma_{\theta}^{-1} = n \left[\text{diag} \left(\theta_{01}^{-1}, \dots, \theta_{0,k-1}^{-1} \right) + p_{0,k}^{-1} \mathbf{1}_{k-1} \mathbf{1}_{k-1}' \right]$$

where $\text{diag} \left(\theta_{01}^{-1}, \dots, \theta_{0,k-1}^{-1} \right)$ denotes the diagonal matrix with the specified diagonal elements. Then we can write

$$(3.10) \quad n \left[\sum_{i=1}^{k-2} \frac{\left(\hat{\theta}_i - \theta_{0i} \right)^2}{\theta_{0i}} + p_{0k}^{-1} \left[\sum_{i=1}^{k-1} \left(\hat{\theta}_i - \theta_{0i} \right) \right]^2 \right]$$

But we can write

$$(3.11) \quad \frac{\left(\hat{\theta}_i - \theta_{0i} \right)^2}{\theta_{0i}} = \frac{\left(\hat{p}_i - p_{0i} \right)^2}{p_{0i}} \quad \text{for} \quad i = 1, \dots, k-1$$

$$\sum_{i=1}^{k-1} \left(\hat{\theta}_i - \theta_{0i} \right) = 1 - \hat{p}_k - (1 - p_{0k}) = -(\hat{p}_k - p_{0k})$$

so that

$$(3.12) \quad W = n \sum_{i=1}^k \frac{\left(\hat{p}_i - p_{0i} \right)^2}{p_{0i}}$$

or

$$(3.13) \quad W = \sum_{i=1}^k \frac{(n_i - np_{0i})^2}{np_{0i}}$$

We derived this form of the statistic by deleting the last category, but it can be shown that deleting any one of the categories yields exactly the same statistic. In this form the Wald statistic W is the same as the Pearson Chi-Square statistic X^2 given by

$$(3.14) \quad X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

$O_i = n_i$ the *Observed* count in the i -th category

$E_i = np_{0i}$ the *Expected* count in the i -th category

The derivation leading up to the representation in (3.13) can be taken as a proof that the Pearson Chi-Square statistic X^2 has a large sample Chi-Square distribution with $k-1$ degrees-of-freedom.

The Likelihood Ratio Statistic G^2

Under the null hypothesis H_0 in (3.1) we can write the likelihood as

$$(3.15) \quad L(\mathbf{p}_0; n_1, \dots, n_k) = \frac{n!}{\prod_{i=1}^k n_i!} \prod_{i=1}^k p_{0i}^{n_i}$$

Under the alternation hypothesis H_A in (3.1) the likelihood function achieves the maximum value given by

$$(3.16) \quad L(\hat{\mathbf{p}}; n_1, \dots, n_k) = \frac{n!}{\prod_{i=1}^k n_i!} \prod_{i=1}^k \hat{p}_i^{n_i} = \frac{n!}{\prod_{i=1}^k n_i!} \prod_{i=1}^k \left(\frac{n_i}{n} \right)^{n_i}$$

Then the likelihood ration statistic Λ is given by

$$(3.17) \quad \Lambda = \frac{L(\mathbf{p}_0; n_1, \dots, n_k)}{L(\hat{\mathbf{p}}; n_1, \dots, n_k)} = \prod_{i=1}^k \left(\frac{np_{0i}}{n_i} \right)^{n_i}$$

In general the statistic $-2 \ln \Lambda$ is asymptotically distributed as a χ^2 random variable with degrees-of-freedom given by the number of free parameters under H_A minus the number of free parameters under H_0 . In this case, this would imply that

$$(3.18) \quad -2 \ln \Lambda = 2 \sum_{i=1}^k n_i \ln \left(\frac{n_i}{np_{0i}} \right) \sim \chi_{k-1}^2 \text{ in large samples.}$$

In categorical data analysis we refer to this as the G^2 statistic, where

$$(3.19) \quad G^2 = -2 \ln \Lambda$$

which in this case can be written as

$$(3.20) \quad G^2 = 2 \sum_{i=1}^k n_i \ln \left(\frac{n_i}{np_{0i}} \right)$$

$$G^2 = 2 \sum_{i=1}^k O_i \ln \left(\frac{O_i}{E_i} \right)$$

where O_i and E_i were defined in (3.14).

Comparison of X^2 and G^2

Theoretically both X^2 and G^2 have an asymptotic χ_{k-1}^2 distribution, but in practice it is often the case that X^2 and G^2 are close computationally. We can show this mathematically by using the Taylor series expansion

$$(3.21) \quad \ln(x) = (x-1) - \frac{1}{2}(x-1)^2 + R(x) \quad \text{for} \quad x \in (0, 2]$$

where $R(x)$ is a remainder term. Then we can write

$$(3.22) \quad G^2 = 2 \sum_{i=1}^k O_i \ln \left(\frac{O_i}{E_i} \right) = -2 \sum_{i=1}^k O_i \ln \left(\frac{E_i}{O_i} \right)$$

$$\begin{aligned}
&= -2 \sum_{i=1}^k O_i \left[\left(\frac{E_i}{O_i} - 1 \right) - \frac{1}{2} \left(\frac{E_i}{O_i} - 1 \right)^2 + R \left(\frac{E_i}{O_i} \right) \right] \\
&= -2 \sum_{i=1}^k \left[(E_i - O_i) - \frac{1}{2} \frac{(O_i - E_i)^2}{O_i} + O_i R \left(\frac{E_i}{O_i} \right) \right] \\
&= \sum_{i=1}^k \frac{(O_i - E_i)^2}{O_i} - 2 \sum_{i=1}^k O_i R \left(\frac{E_i}{O_i} \right) \\
&\approx \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} = X^2
\end{aligned}$$

where replaced O_i by E_i . The approximations we have made hold when $E_i^{-1}O_i \approx 1$ which holds when sample is large and the hypothesis under which E_i is computed is true.

Example 2. In testing crossing two varieties of beans a researcher named Lindstrom found four distinct types of plants in the second generation. In a sample of 1301 plants he found

$$\begin{aligned}
n_1 &= 773 \text{ Green} \\
n_2 &= 231 \text{ Golden} \\
n_3 &= 238 \text{ Green-striped} \\
n_4 &= 59 \text{ Golden-Green-striped}
\end{aligned}$$

According to a simple type of Mendelian inheritance the probabilities of inheriting these four kinds of plans are (9/16,3/16,3/16,1/16) respectively. For this example we have

$$O_1 = 773 \quad E_1 = (1301) \frac{9}{16} = 731.8125$$

$$O_2 = 231 \quad E_2 = (1301) \frac{3}{16} = 243.9375$$

$$O_3 = 238 \quad E_3 = (1301) \frac{3}{16} = 243.9375$$

$$O_4 = 59 \quad E_4 = (1301) \frac{1}{16} = 81.3125$$

We obtain

$$X^2 = 9.27141515$$

$$G^2 = 9.89516144$$

and we look up the values

$$\chi_3^2(0.05) = 7.81$$

$$\chi_3^2(0.025) = 9.35$$

$$\chi_3^2(0.01) = 11.34$$

Therefore both X^2 and G^2 are significant at the 0.05 level but not significant at the 0.01 level. ■